Data Mining - CSC/DSCC 240

Lecture 16: Final Project Guidelines

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Logistics

- **Hope you are well!**
- **Happy Halloween!**
- **Hope your Halloween dresses are well-perceived!**







Today: Final Project

- **Project in a nutshell**
- **What types of projects can I do?**
- **Storytelling process**
- **Statistical tools**
- **Pitfalls to avoid**
- **Examples of good projects are posted.**

Final Project

Different ‘types’ of final projects

Your final final project may have different ***approaches***:

1. Descriptive project

2. Explanatory project

3. Predictive project

✓ **Please refer to the final project guidelines for more information!**

Descriptive Project

Things to keep in mind:

- ✓ Focus on creating a data-story!
- ✓ Ideally use a dataset that not many people has access to
  - ✓ Think about scraping ☺ Data is very important for this project type.
- ✓ No clear causal structure needed
- ✓ Explore the patterns in the dataset
- ✓ Works especially well when the literature is new / non-existent
- ✓ Focus on ‘what’ questions
- ✓ **Typical project examples:** Projects on sharing economy (AirBnb, Uber etc.), recent social/political events and their repercussions on social media, purchasing behavior, marketing…
- ✓ Don’t limit yourself to these topics… ☺
- ✓ Answer ‘what’ questions

Explanatory project

Things to keep in mind:

- ✓ Use existing literature
- ✓ 'Invent a ***puzzle***'
- ✓ You need to have questions / arguments / hypotheses in your paper
- ✓ Think about how you can improve the literature
- ✓ You are welcome to use relatively well-established datasets used in other studies
- ✓ Think about using advanced advanced explanatory / causal inference techniques
- ✓ Think about focusing on quasi-experiments / experiments
- ✓ Your visuals / tables / results / information you share should be strongly connected to the 'story' (e.g. pieces of a puzzle)
- ✓ **Typical example**: Re-visiting an established theory with a new dataset or a new method
- ✓ Answer 'how' questions

Predictive Project

Things to keep in mind:

- ✓ Identify patterns over time
- ✓ Predict what may happen in the future
- ✓ Connect your story to an 'economically-valuable' argument
  - ✓ Example: Because of this 'very accurate prediction', we will be able to reduce the number of birds being killed by wind turbines by 10,000/year which results in …
- ✓ Provide a comparison (usually between 'well-established' and 'new' methods)
  - ✓ Example: Author X can predict this phenomenon with 93.24% accuracy, we are able to do it with 95.46% accuracy …
- ✓ Thorough discussion on the models / methodology
- ✓ Advanced visuals, tables, numerical results should be part of your project

Grading Rubric

Grading Rubric

Grading rubric for the ***prospectus***:

- ✓ Not more than or less than **1-page** (= exactly 1-page)

- ✓ Each section in your prospectus will be given a score (**out of 10**)
  - ✓ Introduction
  - ✓ Data
  - ✓ Literature Review
  - ✓ Methods
  - ✓ Hypothesis / Goals
  - ✓ Expectations about results
  - ✓ **Timeline and Division of Labor**

- ✓ Please refer to guidelines for more information

Grading Rubric

Grading rubric for the ***presentation*** *and* ***paper***:

- ✓ Presentation = ~***15 minutes***
- ✓ Paper = min. ***5 pages*** (in ***IEEE*** format)
- ✓ Each section in your paper will be given a score (**out of 10**)
  - ✓ Introduction
  - ✓ Literature review
  - ✓ Data
  - ✓ Methods ← **You need to include at least one feature engineering, one clustering, and one classification method**
  - ✓ Hypotheses / goals
  - ✓ Results
  - ✓ Conclusion
  - ✓ **Amount of challenge / difficulty** ← **Don't forget ☺**
  - ✓ **Formatting and visual quality**
- ✓ Please refer to written guidelines for more information

Guidelines

Introduction

Please discuss:

- ✓ A summary of what you are trying to do (very briefly):
  - ✓ Example: In this project, we are trying to estimate how education has an impact on future earnings …
- ✓ A summary on what other people have already achieved (very briefly):
  - ✓ Example: The literature provides mixed evidence: education is expected to increase future earnings, but there is also contradicting research
- ✓ A summary on what you have achieved (very briefly):
  - ✓ Example: Using ML techniques, we validate the finding that education improves
- ✓ ***Concise and technical introductions will be rewarded.***

Guidelines

Literature review

- ✓ Find the most relevant literature you can identify and provide a very short overview of:
  - ✓ **1-2** articles for your prospectus
  - ✓ **5-10** articles for your presentation / paper
  - ✓ Note: Most relevant does not mean most highly cited

- ✓ The articles you study should be very closely related to your Project idea
  - ✓ Note: Talking about similar topics does not (necessarily) mean that they are very closely related
  - ✓ Make sure that the ind. variables and dep. variable in the articles you find are very similar / the same
- ✓ ***The use of related and respected articles will be rewarded.***

Guidelines

Data / Descriptive Analysis

Please discuss:

- ✓ Where you obtained the data
  - ✓ References, code for scraping, permission to use the data …
  - ✓ If you used scraping tools/code, please talk about your efforts
- ✓ Introduce the data to your reader
  - ✓ Advanced visuals, tables, summary figures, explanations
  - ✓ Size of your data
  - ✓ How it compares to other similar datasets
  - ✓ Why you chose to use this dataset
  - ✓ All papers (descriptive, explanatory, predictive) need to have a descriptive analysis section
- ✓ ***Greater efforts to collect data and advanced descriptive analyses will be rewarded.***

Guidelines

Goals / Hypotheses

Please discuss:

- ✓ What you want to achieve with this project
  - ✓ Example: We want to improve on the strength of the causal connection between education and future earnings by using a new dataset (i), a new algorithm (ii), or both (iii)
- ✓ Provide a discussion on the methods you will be using
- ✓ Provide measurable and falsifiable questions with a direction:
  - ✓ Example: Does primary education improve aggregated future earnings?
  - X Not: There is a connection between education and future earnings?
  - X Not: Will better education help us to do space travel by 2050?

- ✓ ***Parsimonious hypotheses / goals that are easy to understand+measure+falsify, use of advanced methods will be rewarded.***

Guidelines

Results and Discussion

Please discuss:

- ✓ Your findings in a structured and organized manner
- ✓ Use of bullet points might be of consideration (don't use them very generously)
- ✓ Provide numerical comparisons / findings
  - ✓ Example: We have found out that an extra year of college education improves future earnings by 15,000 USD / year for the alumni of … school
- ✓ You are welcome to provide a table for your findings
- ✓ Please connect your findings to the story
- ✓ Please evaluate on the significance of your findings

- ✓ ***Well-thought, well-written, organized discussions will be rewarded.***

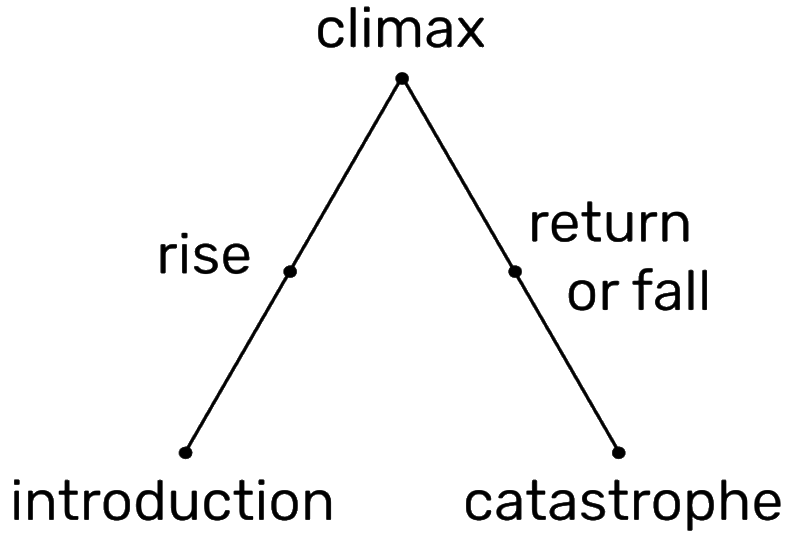
Storytelling

Structuring Your Story

- Most narratives follow a common format in terms of structure
- Freytag's Pyramid
  - A structure of a dramatic work such as a play or film
  - You can choose to **think like a scenarist who is designing a movie for Hollywood!**

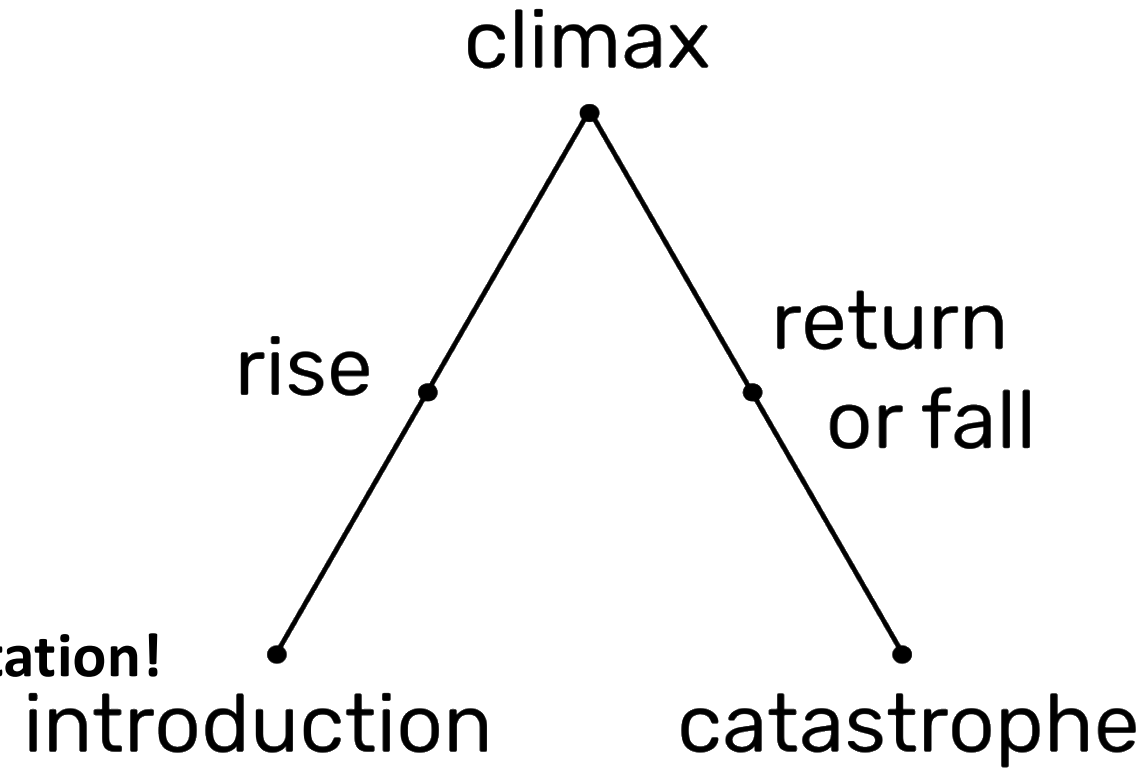


Gustav Freytag



Freytag's Pyramid

- **Exposition**
- **Inciting incident**
- **Complication**
- **Climax**
- **Reversal**
- **Resolution**
- **Denouement**



- **Especially important for the presentation!**
- Let's try to apply this to data science!

Freytag's Pyramid and Statistics

- Provide the audience a few sources of motivation to make them interested (**exposition**)
- Mention a key issue you found to lead your audience into believing that you need data to solve the problem (**inciting incident**)

- Introduce the dataset (**complication**)

- Ask some really interesting questions that go beyond the variables in the dataset (**climax**)
- First provide a few visualizations about your dataset (**reversal**)

- Answer the questions you asked using the data (**resolution**)

- Conclude and tell the significance of your findings (**denouement**)

How can I find stories in data?

- Do a correlation analysis / create a ***correlation table***. What seems to be related?
- Look at the distributions of your variables? Are there any ***outliers***? Is there anything interesting about the ***outliers***?
- Try to identify one key variable, one variable that can be predicted by the other variables in your dataset
  - This variable may become your ***outcome variable***!
- Are there any clear '***groups***' of data points in your dataset? Are there categorical variables? Focus on comparing them.
  - Example: Soccer teams in different leagues
  - Example: Liberal arts colleges vs. big public universities

Creating Context

- A powerful data story allows the audience to feel like they are there, and living the situation
  - **Use pictures, visuals, quotes, time-relevant information to create a story**
  - Examples:
    - Analyzing a dataset on Black Friday shopping season during Black Friday
    - Connecting a dataset on politics to the recent presidential campaigns
    - Telling a data-analytical story about colleges to college students
  - Ask yourself who the audience is and what they need to know!
  - Try to go outside of your comfort zone ☺
  - Try to find a context that people can associate themselves with

Timeline

- Most good stories have a '***linear timeline***':
  - A clear beginning
  - A clear analysis
  - A clear conclusion

- In data language:
  - Data
  - Hypothesis
  - Methods
  - Analysis
  - Findings and Conclusion

Make Your Story Relatable

- Again, there may be 'different' kinds of **final projects**:
  1. Talking about a 'new' topic
     - Example: Analyzing the behavior of autonomous cars
  2. Re-visiting an old topic
     - Example: Analyzing the relationship between trade and peace

For both cases, give an example from today so that the people in the audience can relate themselves!

Focusing on people affected by data rather than the numbers themselves is a better idea!

- Don't forget that you are telling a 'data-story' that impacts people!

Tools



Tools from our class

- You can check the topics in the syllabus ☺

- Supervised learning
- Unsupervised learning
- Dimensionality reduction

- You are encouraged to explore other methods!

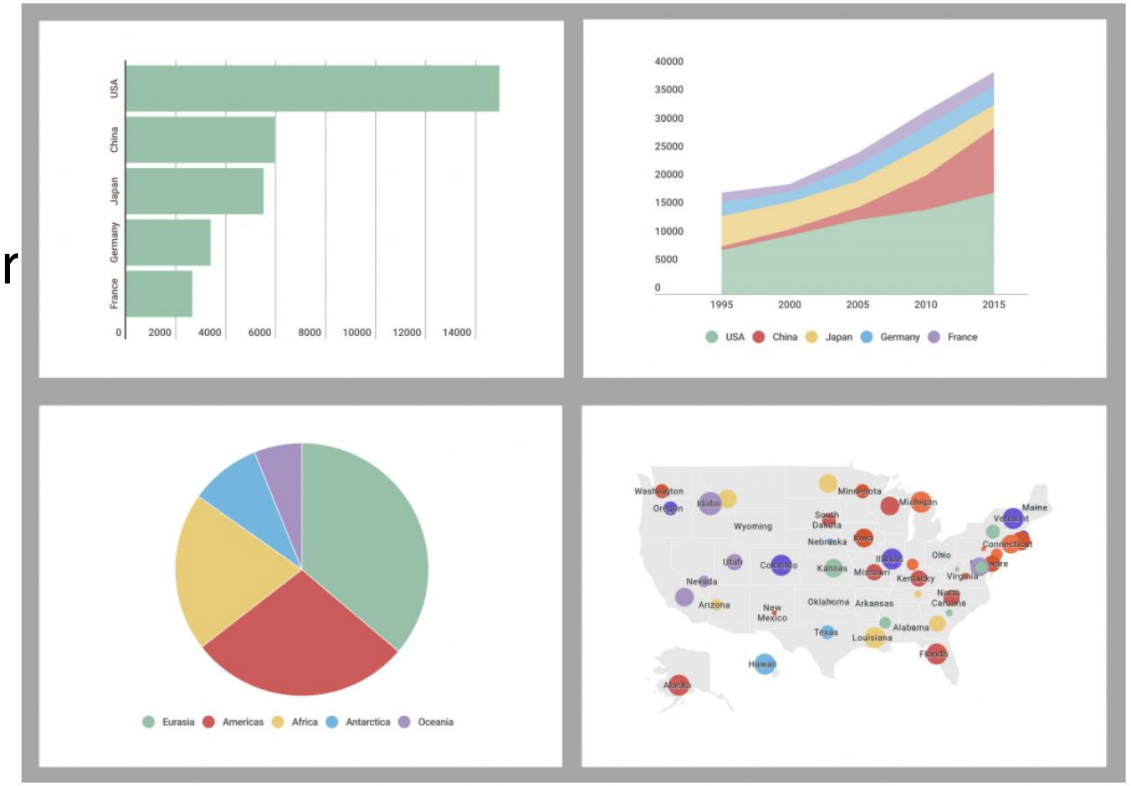
- ***And your efforts to go beyond our toolbox will be rewarded/***
- ***Also check: Good visualization practices we have previously discussed***

Ideally, the technical part of your project is a combination of a lot of advanced techniques!


Graphics

Using Visuals

- Make sure that you use a variety of charts!
- Providing descriptive analysis about your dataset is a good idea for the beginning of your analysis
- **Not the middle or the end of your analysis!**
- Make sure that you strengthen different arguments by the visuals you create



Using Visuals

- If you are ranking quantifiable things:
  - Sort them!
- If you are comparing similar things:
  - Give them the same color!
- Ideally, don't use a lot of decoration:
  - But, you don't need to be as conservative as Edward Tufte!
- If there are many things to compare:
  - Limit the number of things to show to a plausible amount!



- You can also use icons, logos, and pictures, if they are relevant!



Tables and Graphs

- When should you use a table? When should you use a graph?

- A lot of columns, comparably few observations
  - Graphs may do a better job!

- Not so many columns, a lot of observations
  - Tables may do a better job!

- Precision of your description not so important
  - Graphs may do a better job!

- Meaningful absolute quantities
  - Tables may do a good job (sometimes, graphs, as well)!

An example for a very basic explanatory project

Let’s say you are interested in basketball!

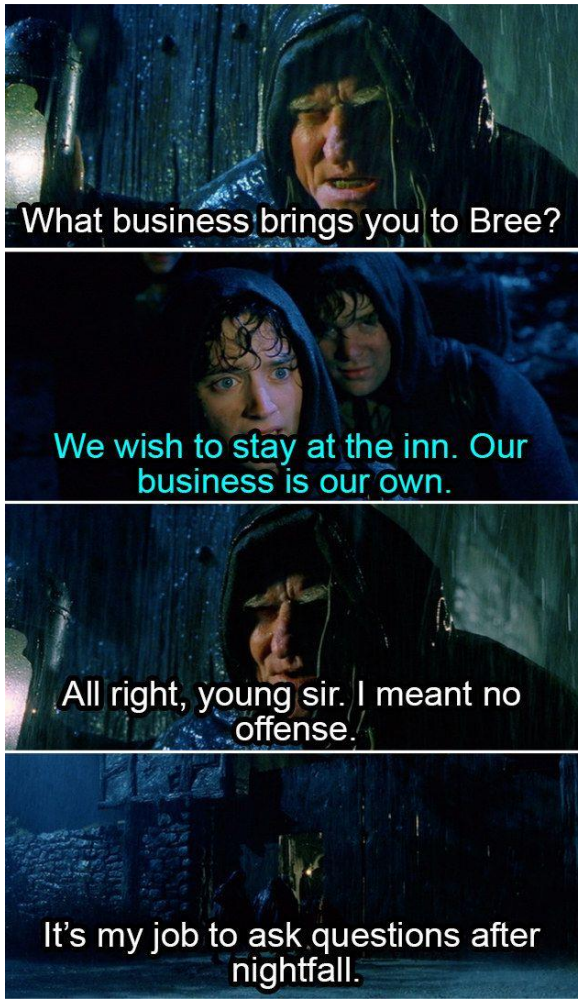


Asking good questions

1) Don’t use numbers in your hypotheses:

- We expect that basketball players taller than 6’6’’ are more successful

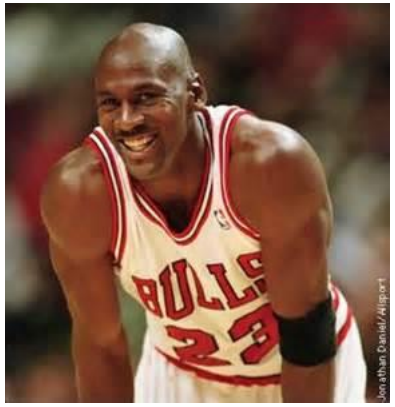
- Ideally:
  - We expect that taller basketball players are more successful
  - Test: Basketball players taller than 6’6’’ are more successful
  - Test: The average height of basketball players in better teams is greater



Asking good questions

2) Don't ask a lot of similar questions:

- We expect taller basketball players to be more successful
- We expect better teams to have taller basketball players
- We expect basketball teams in the USA to have taller basketball players



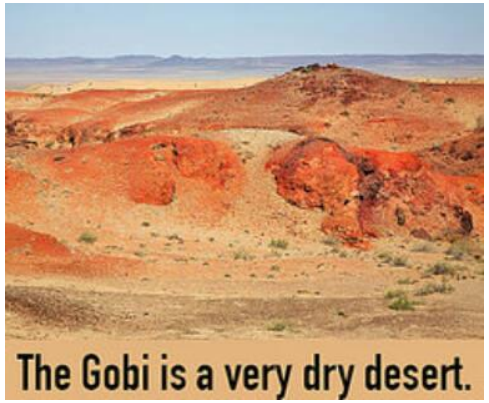
- **<u>Ideally</u>**:
  - We expect taller basketball players to be more successful
  - We also expect that good trainers have a big impact on the success of a basketball player
  - We also expect that age is really important for the success



Asking good questions

2) Avoid tautology. Some examples:

- I will examine the relationship between number of wins and number of championships
- We will look at the relationship between good offensive play and good overall play
- We will analyze the success by looking at points scored



- **<u>Ideally</u>**:
  - We want to look at strategies and number of wins
  - We want to analyze the difference a coach has made
  - We want to examine and compare the games w/ and w/out a player/some players

Asking good questions

4) Number your questions

- Hypotheses are usually numbered as ***H1***, ***H2***, ***H3*** …
- It is a good idea to list your hypotheses or give them numbers (i, ii, iii ; 1, 2, 3 etc.)



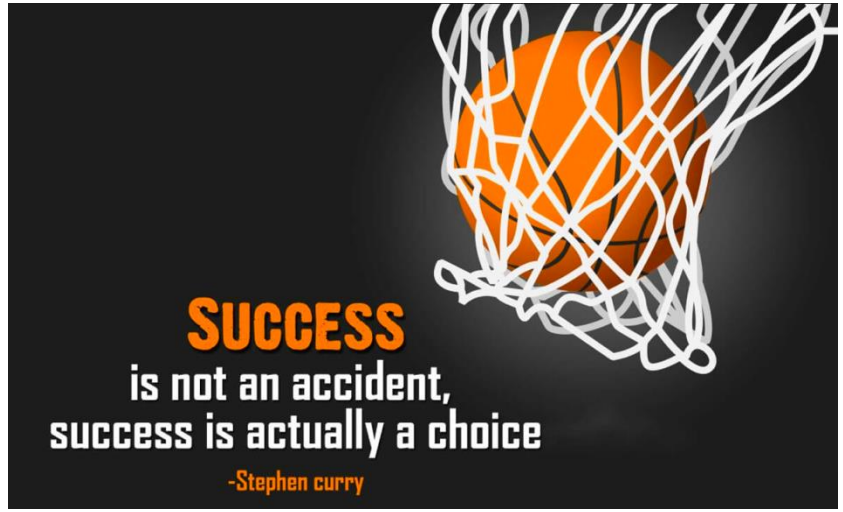
5) Not too many, and not too few questions

- Ideally, 5-7 questions you can explore in depth
- And you can group under 2-3 major topics

Grouping ideas

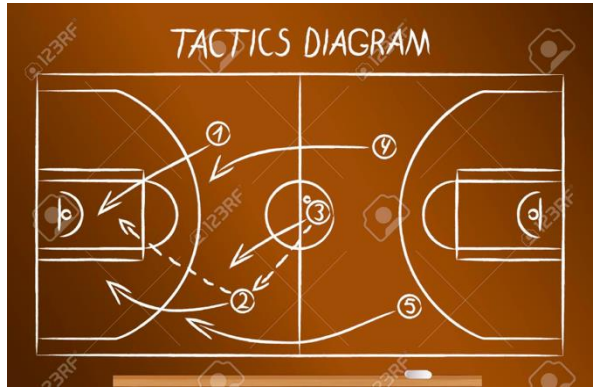
- Try to cover different aspects of the story:
  - Basketball players
  - Trainers
  - Previous success of the team
  - Equipment technology

- So, ideally, don't:
  - Impact of offensive play
  - Impact of defensive play
  - Performance of point guards
  - Performance of shooting guards



Grouping ideas

- Ideally have a few topics and 1-2 related questions:
  - Basketball players
    - Taller players are more successful
    - Mid-career players are more successful
  - Trainers
    - Higher-paid trainers are more successful
  - Previous success of the team
    - Teams who have been undefeated for five games consecutively are more successful
    - Teams who have won the championship at least once are more successful



Methodology

- Again:
  - You can group the techniques
  - You can use a more simple and a more advanced technique for the same question (for instance, linear regression + SVM-regression)

  - Clearly indicate in your paper what you are willing to use!



Concluding your story

- Summarize your findings
  - Talk about numbers
  - Talk about economic significance
  - Why do you think what you found is important?
- Talk about what you think was needed
  - Not enough data (?)
  - Different data collection is needed (?)
  - Do you need more statistical techniques (?)
  - Criticizing yourself is ok, too!
- Connect your story
  - Learning from the past
  - Future implications



Deadlines

- Find your **team**
  - Hopefully, you already found a good team!
- Submit your 1-page prospectus
  - 3% of your overall grade
  - **Deadline: November 15 (Sunday)**
- Presentation
  - 7% of your overall grade
  - **Deadline: November 29 (Sunday)**
- Submit your final paper:
  - 10% of your overall grade
  - **Deadline: December 13 (Sunday)**